

Содержание

Описание лабораторной работы.....	1
Постановка задачи.....	1
Цель и основные этапы работы.....	1
Шаг 1. Подготовка среды (VPC/подсети/SG).....	2
Шаг 2. Развертывание Amazon RDS.....	4
Шаг 3. Создание виртуальной машины для подключения к базе данных.....	7
Шаг 4. Подключение к базе данных и выполнение базовых операций.....	9
Шаг 5. Создание Read Replica.....	12
Шаг 6. Подключение приложения к базе данных.....	15
Вывод.....	16
Веблиография.....	17

Описание лабораторной работы

Постановка задачи

Необходимо будет развернуть и настроить облачную базу данных в Amazon Web Services с использованием сервисов Amazon RDS и Amazon DynamoDB. В рамках работы потребуется создать сетевую инфраструктуру, настроить группы безопасности и подключение между EC2-инстансом и базой данных.

Также будет необходимо создать экземпляр базы данных MySQL в Amazon RDS, подключиться к нему с виртуальной машины EC2 и выполнить базовые операции с данными: создание таблиц, добавление записей и выполнение SQL-запросов. Дополнительно потребуется настроить Read Replica, проверить её работу и изучить особенности репликации данных в облачной среде.

На заключительном этапе будет выполнено подключение веб-приложения к базе данных Amazon RDS и проверка корректной работы CRUD-операций через облачную базу данных.

Цель и основные этапы работы

Целью лабораторной работы является изучение сервисов Amazon RDS и Amazon DynamoDB, а также получение практических навыков

работы с облачными базами данных в AWS. В ходе выполнения работы необходимо освоить создание и настройку экземпляров реляционных баз данных, подключение к ним через EC2-инстанс и выполнение базовых операций с данными.

Основными этапами работы являются подготовка сетевой инфраструктуры и групп безопасности, создание и настройка экземпляра Amazon RDS MySQL, подключение к базе данных с EC2-инстанса, создание таблиц и выполнение SQL-запросов. Также в рамках работы необходимо изучить механизм Read Replica, проверить репликацию данных и подключить приложение к облачной базе данных Amazon RDS.

Шаг 1. Подготовка среды (VPC/подсети/SG)

В этот раз при создании VCP выбираю “VPC and more”

Это автоматически создаст:

- VPC
- public subnet
- private subnet
- route tables
- IGW
- NAT Gateway

A VPC is an isolated portion of the AWS Cloud populated by AWS objects, such as Amazon EC2 instances. Mouse over a resource to highlight the related resources.

VPC settings

Resources to create [Info](#)
Create only the VPC resource or the VPC and other networking resources.

☐ VPC only ☒ VPC and more

Name tag auto-generation [Info](#)
Enter a value for the Name tag. This value will be used to auto-generate Name tags for all resources in the VPC.

☒ Auto-generate

project

IPv4 CIDR block [Info](#)
Determine the starting IP and the size of your VPC using CIDR notation.

10.0.0.0/16 65,536 IPs

CIDR block size must be between /16 and /28.

IPv6 CIDR block [Info](#)
☒ No IPv6 CIDR block
☐ Amazon-provided IPv6 CIDR block

Tenancy [Info](#)
Default

► **Encryption settings - optional**

Number of Availability Zones (AZs) [Info](#)
Choose the number of AZs in which to provision subnets. We recommend at least two AZs for high availability.

1 2 3

► **Customize AZs**

Preview

VPC [Show details](#)
Your AWS virtual network

project-vpc

Subnets (4)
Subnets within this VPC

eu-central-1a

- project-subnet-public1-eu-central-
- project-subnet-private1-eu-central-

eu-central-1b

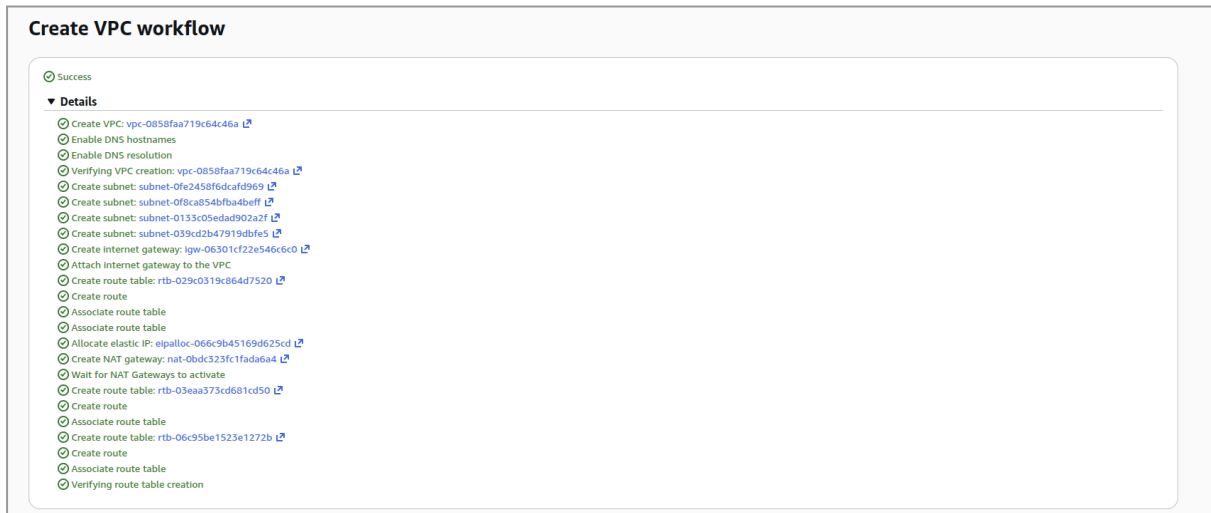
- project-subnet-public2-eu-central-
- project-subnet-private2-eu-central-

Route tables (3)
Route network traffic to resources

- project-rtb-public
- project-rtb-private1-eu-central-
- project-rtb-private2-eu-central-

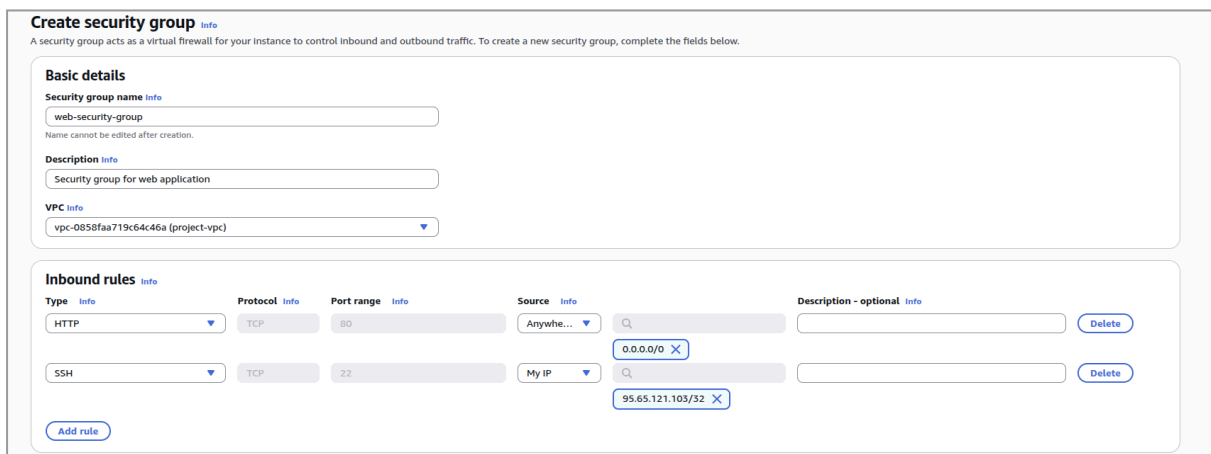
AWS автоматически создаст:

- VPC: project-vpc
- Public subnets: project-subnet-public1 project-subnet-public2
- Private subnets: project-subnet-private1 project-subnet-private2
- Internet Gateway: project-igw
- NAT Gateway project-nat:
- Route Tables: public и private.



Создала web-security-group с правилами входящего трафика:

- HTTP (порт 80) от любого источника;
- SSH (порт 22) от вашего IP-адреса или от любого источника (для учебных целей);



Создала db-mysql-security-group с правилами входящего трафика:

- MySQL/Aurora (порт 3306) от web-security-group

Create security group [Info](#)

A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. To create a new security group, complete the fields below.

Basic details

Security group name [Info](#)

db-mysql-security-group

Name cannot be edited after creation.

Description [Info](#)

Security group for MySQL

VPC [Info](#)

vpc-0858faa719c64c46a [project-vpc]

Inbound rules [Info](#)

Type	Protocol	Port range	Source	Description - optional	
MySQL/Aurora	TCP	3306	Custom	sg-0710c53afc7efc849	Delete
				sg-0710c53afc7efc849	X

[Add rule](#)

Изменила web-security-group, добавив правило для исходящего трафика:

- Исходящий: MySQL/Aurora (порт 3306) к db-mysql-security-group

Edit outbound rules [Info](#)

Outbound rules control the outgoing traffic that's allowed to leave the instance.

Outbound rules [Info](#)

Security group rule ID: sgr-0847b8747da8db82e

Type	Protocol	Port range	Destination	Description - optional	
All traffic	All	All	Custom		Delete
			0.0.0.0/0		X
MySQL/Aurora	TCP	3306	Custom	sg-0fb607c6b4a5440a3	Delete
			sg-0fb607c6b4a5440a3		X

[Add rule](#)

[Cancel](#) [Preview changes](#) [Save rules](#)

Шаг 2. Развертывание Amazon RDS

Создала Subnet Group:

Subnet group details

Name

You won't be able to modify the name after your subnet group has been created.

project-rds-subnet-group

Must contain from 1 to 255 characters. Alphanumeric characters, spaces, hyphens, underscores, and periods are allowed.

Description

Subnet group for RDS

VPC

Choose a VPC identifier that corresponds to the subnets you want to use for your DB subnet group. You won't be able to choose a different VPC identifier after your subnet group has been created.

project-vpc (vpc-0858faa719c64c46a)

4 Subnets, 2 Availability Zones

Add subnets

Availability Zones

Choose the Availability Zones that include the subnets you want to add.

Choose an availability zone

eu-central-1a [X](#) eu-central-1b [X](#)

Subnets

Choose the subnets that you want to add. The list includes the subnets in the selected Availability Zones.

Select subnets

project-subnet-private1-eu-central-1a [X](#) project-subnet-private2-eu-central-1b [X](#)

Subnet ID: subnet-0133c05eda9902a2f CIDR: 10.0.128.0/20 Subnet ID: subnet-039cd2b47919dbfe5 CIDR: 10.0.144.0/20

[i](#) For Multi-AZ DB clusters, you must select 3 subnets in 3 different Availability Zones.

Вопрос: Что такое Subnet Group? И зачем необходимо создавать Subnet Group для базы данных?

Ответ: Subnet Group - это набор подсетей, в которых AWS может разместить базу данных RDS.

Она нужна, чтобы база данных запускалась внутри нужной VPC, AWS мог разместить БД в разных Availability Zones, база данных находилась именно в приватных подсетях, а не в публичных.

Создала экземпляр базы данных:

Engine options

Engine type [Info](#)

☒ Aurora (MySQL Compatible)

☐ Aurora (PostgreSQL Compatible)

☒ MySQL

☐ PostgreSQL

☐ MariaDB

☐ Oracle

☐ Microsoft SQL Server

☐ IBM Db2

Choose a database creation method

☐ Full configuration

☒ Easy create

Configuration

Edition

☒ MySQL Community

DB instance size

☐ Production

☐ Dev/Test

☒ Free tier

DB instance identifier

Type a name for your DB instance. The name must be unique across all DB instances owned by your AWS account in the current AWS Region.

project-rds-mysql-prod

The DB instance identifier is case-insensitive, but is stored as all lowercase (as in "mydbinstance"). Constraints: 1 to 63 alphanumeric characters or hyphens. First character must be a letter. Can't contain two consecutive hyphens. Can't end with a hyphen.

Master username [Info](#)

Type a login ID for the master user of your DB instance.

admin

1 to 16 alphanumeric characters. The first character must be a letter.

Credentials management

You can use AWS Secrets Manager or manage your master user credentials.

☐ Managed in AWS Secrets Manager - most secure

☒ Self managed

☐ Auto generate password

Amazon RDS can generate a password for you, or you can specify your own password.

Master password [Info](#)

Password strength [Strong](#)

Minimum constraints: At least 8 printable ASCII characters. Can't contain any of the following symbols: / ' * @

Confirm master password [Info](#)

Настройка инстанса:

Instance configuration

The DB instance configuration options below are limited to those supported by the engine that you selected above.

DB instance class [Info](#)

▼ Hide filters

☒ Show instance classes that support Amazon RDS Optimized Writes [Info](#)

☒ Include previous generation classes

☐ Standard classes (includes m classes)

☐ Memory optimized classes (includes r and x classes)

☒ Burstable classes (includes t classes)

Instance type

db.t3.micro

2 vCPUs 1 GiB RAM EBS Bandwidth: Up to 2,085 Mbps Network: Up to 5 Gbps

Настройка Storage:

Storage

Storage type [Info](#)

Provisioned IOPS SSD (io2) storage volumes are now available.

General Purpose SSD (gp3)
Performance scales independently from storage

Allocated storage [Info](#)

20

GIB

Minimum: 20 GiB. Maximum: 6,144 GiB

Provisioned IOPS [Info](#)

3000

IOPS

Baseline IOPS of 3,000 IOPS is included for allocated storage less than 400 GiB.

Storage throughput [Info](#)

125

MIBps

Baseline storage throughput of 125 MiBps is included for allocated storage less than 400 GiB.

To provision additional IOPS and throughput, increase the allocated storage to 400 GiB or greater.

Additional storage configuration

Storage autoscaling [Info](#)

Provides dynamic scaling support for your database's storage based on your application's needs.

☒ Enable storage autoscaling
Enabling this feature will allow the storage to increase after the specified threshold is exceeded.

Maximum storage threshold [Info](#)

Charges will apply when your database autoscales to the specified threshold

100

GIB

Allocated storage value must be 22 GiB to 6,144 GiB

Раздел Connectivity:

Connectivity [Info](#)

Compute resource

Choose whether to set up a connection to a compute resource for this database. Setting up a connection will automatically change connectivity settings so that the compute resource can connect to this database.

☒ Don't connect to an EC2 compute resource
Don't set up a connection to a compute resource for this database. You can manually set up a connection to a compute resource later.

☐ Connect to an EC2 compute resource
Set up a connection to an EC2 compute resource for this database.

Network type [Info](#)

To use dual-stack mode, make sure that you associate an IPv6 CIDR block with a subnet in the VPC you specify.

☒ IPv4
Your resources can communicate only over the IPv4 addressing protocol.

☐ Dual-stack mode
Your resources can communicate over IPv4, IPv6, or both.

Virtual private cloud (VPC) [Info](#)

Choose the VPC. The VPC defines the virtual networking environment for this DB instance.

project-vpc (vpc-0858faa719c64c46a)
4 Subnets, 2 Availability Zones

After a database is created, you can't change its VPC. Only VPCs with a corresponding DB subnet group are listed.

After a database is created, you can't change its VPC.

DB subnet group [Info](#)

Choose the DB subnet group. The DB subnet group defines which subnets and IP ranges the DB instance can use in the VPC that you selected.

project-rds-subnet-group
2 Subnets, 2 Availability Zones

Public access [Info](#)

☐ Yes
RDS assigns a public IP address to the database. Amazon EC2 instances and other resources outside of the VPC can connect to your database. Resources inside the VPC can also connect to the database. Choose one or more VPC security groups that specify which resources can connect to the database.

☒ No
RDS doesn't assign a public IP address to the database. Only Amazon EC2 instances and other resources inside the VPC can connect to your database. Choose one or more VPC security groups that specify which resources can connect to the database.

VPC security group (firewall) [Info](#)

Choose one or more VPC security groups to allow access to your database. Make sure that the security group rules allow the appropriate incoming traffic.

Дополнительно:

Option group [Info](#)
default:mysql-8-4

Backup
☒ **Enable automated backup**
Creates a point-in-time snapshot of your database.

Backup retention period [Info](#)
The number of days (1-35) for which automatic backups are kept.
1 day

Backup window [Info](#)
The daily time range (in UTC) during which RDS takes automated backups.
☐ Choose a window
☒ No preference

Backup tags [Info](#)
Copies tags from the source database to the automated backup and snapshots respectively.
☐ Copy tags to automated backup
This is a one-time setting. Future tag modifications need manual updates.
☒ Copy tags to snapshots

Backup replication [Info](#)
☐ Enable replication in another AWS Region
Enabling replication automatically creates backups of your DB Instance in the selected Region, for disaster recovery, in addition to the current Region.
☐ Enable encryption
Choose to encrypt the given instance. Master key IDs and aliases appear in the list after they have been created using the AWS Key Management Service console. [Info](#)

Maintenance
Auto minor version upgrade [Info](#)
☐ Enable auto minor version upgrade
Enabling auto minor version upgrade will automatically upgrade your database minor version. For limitations and more details, see [Automatically upgrading the minor engine version documentation](#).

Создание базы успешно завершилось:

Databases (1) Group resources Modify

Filter by databases

DB identifier	Status	Role	Engine	Upgrade rollout order	Region ...	Size
project-rds-mysql-prod2	Available	Instance	MySQL Co...	SECOND	eu-central...	db.t3.micro

Endpoint для подключения:
http://project-rds-mysql-prod2.clawcms0eqa7.eu-central-1.rds.amazonaws.com

Connectivity & security | Monitoring | Logs & events | Configuration | Zero-ETL integrations | Maintenance & backups | Data migrations | Tags | Recommendations

Connect using [Info](#)

☒ **Code snippets**
Use when connecting through SDK, APIs, or third-party tools including agents.

☐ **CloudShell**
Use for a quick access to AWS CLI that launches directly from the AWS Management Console.

☐ **Endpoints**
Use when connecting through any IDE interface.

Internet access gateway
☐ Disabled

IAM Authentication
☐ Disabled

Programming language MySQL (macOS) **Endpoint type** Instance endpoint

Connection steps
Follow the steps below to paste the code of each step in your tool and run the commands. The snippets dynamically reflect the authentication configuration.

```
1 curl -o global-bundle.pem https://truststore.pki.rds.amazonaws.com/global/global-bundle.pem
```

```
1 mysql -h project-rds-mysql-prod2.clawcms0eqa7.eu-central-1.rds.amazonaws.com -P 3306 -u admin -p --ssl-mode=VERIFY_IDENTITY --ssl-ca=./global-bundle.pem
```

Шаг 3. Создание виртуальной машины для подключения к базе данных

Создала:

[EC2](#) > [Instances](#) > Launch an instance

Launch an instance Info

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags Info

Name

[Add additional tags](#)

▼ Application and OS Images (Amazon Machine Image) Info

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Recents

Quick Start

Amazon Linux

aws

macOS

Mac

Ubuntu

Canonical Ubuntu

Windows

Microsoft

Red Hat

Red Hat

SUSE Linux

SUSE

Debian

debian

[Browse more AMIs](#)
 Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

Amazon Linux 2023 kernel-6.1 AMI

ami-036bdae36143a955f (64-bit (x86), uefi-preferred) / ami-0abb1c87d6e8729b7 (64-bit (Arm), uefi)
 Virtualization: hvm ENA enabled: true Root device type: ebs

Free tier eligible

Description

Amazon Linux 2023 (kernel-6.1) is a modern, general purpose Linux-based OS that comes with 5 years of long term support. It is optimized for AWS and designed to provide a secure, stable and high-performance execution environment to develop and run your cloud applications.

Amazon Linux 2023 AMI 2023.11.20260514.0 x86_64 HVM kernel-6.1

Раздел Network settings:

▼ Network settings Info

VPC - required Info

10.0.0.0/16

Subnet Info

project-subnet-public1-eu-central-1a

VPC: vpc-0858faa719c64c46a Owner: 523271873433 Availability Zone: eu-central-1a (euc1-az2)
 Zone type: Availability Zone IP addresses available: 4090 CIDR: 10.0.0.0/20

Auto-assign public IP Info

Firewall (security groups) Info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☐ Create security group
 ☒ Select existing security group

Common security groups Info

VPC: vpc-0858faa719c64c46a

[Compare security group rules](#)

Security groups that you add or remove here will be added to or removed from all your network interfaces.

► Advanced network configuration

Раздел User data:


```
[ec2-user@ip-10-0-6-244 ~]$ mysql -h project-rds-mysql-prod2.clawcms0eqa7.eu-central-1.rds.amazonaws.com -P 3306 -u admin -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MySQL connection id is 34
Server version: 8.4.9 Source distribution

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MySQL [(none)]> 
```

Создала две таблицы:

```
CREATE TABLE categories (
    id INT AUTO_INCREMENT PRIMARY KEY,
    name VARCHAR(100) NOT NULL
);

CREATE TABLE todos (
    id INT AUTO_INCREMENT PRIMARY KEY,
    title VARCHAR(255) NOT NULL,
    status VARCHAR(50),
    category_id INT,
    FOREIGN KEY (category_id) REFERENCES categories(id)
);
```

Вставила записи в таблицы:

```
INSERT INTO categories (name)
VALUES
('Study'),
('Work'),
('Personal');

INSERT INTO todos (title, status, category_id)
VALUES
('Finish AWS lab', 'in progress', 1),
('Prepare report', 'done', 1),
('Send email', 'pending', 2);
```

Проверила результат:

```

MySQL [project_db]> SELECT * FROM categories;
+----+-----+
| id | name   |
+----+-----+
|  1 | Study  |
|  2 | Work   |
|  3 | Personal |
+----+-----+
3 rows in set (0.002 sec)

MySQL [project_db]> SELECT * FROM todos;
+----+-----+-----+-----+
| id | title          | status      | category_id |
+----+-----+-----+-----+
|  1 | Finish AWS lab | in progress | 1           |
|  2 | Prepare report | done        | 1           |
|  3 | Send email     | pending     | 2           |
+----+-----+-----+-----+
3 rows in set (0.001 sec)

MySQL [project_db]> 

```

Все верно заполнилось

Делаю JOIN на таблицы:

```

SELECT
    todos.id,
    todos.title,
    todos.status,
    categories.name AS category
FROM todos
JOIN categories
ON todos.category_id = categories.id;

```

```

MySQL [project_db]> SELECT
->     todos.id,
->     todos.title,
->     todos.status,
->     categories.name AS category
-> FROM todos
-> JOIN categories
-> ON todos.category_id = categories.id;
+----+-----+-----+-----+
| id | title          | status      | category |
+----+-----+-----+-----+
|  1 | Finish AWS lab | in progress | Study    |
|  2 | Prepare report | done        | Study    |
|  3 | Send email     | pending     | Work     |
+----+-----+-----+-----+
3 rows in set (0.001 sec)

```

CRUD операции:

```
MySQL [project_db]> UPDATE todos
  -> SET status = 'done'
  -> WHERE id = 1;
Query OK, 1 row affected (0.008 sec)
Rows matched: 1  Changed: 1  Warnings: 0

MySQL [project_db]> DELETE FROM todos
  -> WHERE id = 3;
Query OK, 1 row affected (0.005 sec)

MySQL [project_db]> SELECT * FROM todos;
+----+-----+-----+-----+
| id | title          | status | category_id |
+----+-----+-----+-----+
|  1 | Finish AWS lab | done   |            1 |
|  2 | Prepare report | done   |            1 |
+----+-----+-----+-----+
2 rows in set (0.001 sec)
```

Шаг 5. Создание Read Replica

Настроила по требованиям:

Connectivity

Network type Info
 To use dual-stack mode, make sure that you associate an IPv6 CIDR block with a subnet in the VPC you specify.

☒ **IPv4**
 Your resources can communicate only over the IPv4 addressing protocol.

☐ **Dual-stack mode**
 Your resources can communicate over IPv4, IPv6, or both.

DB subnet group Info
 Choose the DB subnet group. The DB subnet group defines which subnets and IP ranges the DB instance can use in the VPC that you selected.

project-rds-subnet-group

Public access
☐ **Publicly accessible**
 RDS assigns a public IP address to the database. Amazon EC2 instances and other resources outside of the VPC can connect to your database. Resources inside the VPC can also connect to the database. Choose one or more VPC security groups that specify which resources can connect to the database.
 ☒ **Not publicly accessible**
 No IP address is assigned to the DB instance. EC2 instances and devices outside the VPC can't connect.

Existing VPC security groups
 Choose VPC security groups

default
 db-mysql-security-group

Availability Zone Info
 The EC2 Availability Zone that the database will be created in.

No preference

Certificate authority - optional Info
 Using a server certificate provides an extra layer of security by validating that the connection is being made to an Amazon database. It does so by checking the server certificate that is automatically installed on all databases that you provision.

rds-ca-rsa2048-g1 (default)
 Expiry: May 22, 2061

 If you don't select a certificate authority, RDS chooses one for you.

Additional configuration

Дождалась успешного запуска:

Successfully created replica [project-rds-mysql-read-replica](#) in Europe (Frankfurt)

Databases (2) Group resources Mod

Filter by databases

	DB identifier	Status	Role	Engine	Upgrade rollout order	Region ...	Size
☐	project-rds-mysql-prod2	Available	Primary	MySQL Co...	SECOND	eu-central...	db.t3.micro
☐	project-rds-mysql-read-replica	Available	Replica	MySQL Co...	SECOND	eu-central...	db.t3.micro

Вошла с консоли:

```
[ec2-user@ip-10-0-6-244 ~]$ mysql -h project-rds-mysql-read-replica.clawcms0eqa7
.eu-central-1.rds.amazonaws.com -P 3306 -u admin -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MySQL connection id is 32
Server version: 8.4.9 Source distribution

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MySQL [(none)]> USE project_db;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
MySQL [project_db]>
```

Протестировала запросы:

```
MySQL [project_db]> SELECT * FROM categories;
+----+-----+
| id | name  |
+----+-----+
| 1  | Study |
| 2  | Work  |
| 3  | Personal |
+----+-----+
3 rows in set (0.005 sec)

MySQL [project_db]> SELECT * FROM todos;
+----+-----+-----+-----+
| id | title          | status | category_id |
+----+-----+-----+-----+
| 1  | Finish AWS lab | done   | 1           |
| 2  | Prepare report | done   | 1           |
+----+-----+-----+-----+
2 rows in set (0.004 sec)
```

Вопрос: Какие данные вы видите? Объясните почему.

Ответ: Вижу те же данные, что и на основной базе данных. Потому что Read Replica автоматически копирует данные с основного экземпляра RDS.

Тестирую Insert:

```
MySQL [project_db]> INSERT INTO categories(name)
-> VALUES ('Test');
ERROR 1290 (HY000): The MySQL server is running with the --read-only option so it cannot execute this statement
MySQL [project_db]> 
```

Вопрос: Получилось ли выполнить запись на Read Replica? Почему?

Ответ: Запись не выполнялась, Read Replica предназначена только для чтения. AWS блокирует: INSERT, UPDATE, DELETE, потому что реплика синхронизируется с основной базой и не должна изменяться отдельно.

```
MySQL [(none)]> USE project_db;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
MySQL [project_db]> INSERT INTO categories(name)
-> VALUES ('AWS');
Query OK, 1 row affected (0.021 sec)

MySQL [project_db]> exit
Bye
[ec2-user@ip-10-0-6-244 ~]$ mysql -h project-rds-mysql-read-replica.clawcms0eqa7.eu-central-1.rds.amazonaws.com -P 3306 -u admin -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MySQL connection id is 38
Server version: 8.4.9 Source distribution

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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MySQL [(none)]> USE project_db;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
MySQL [project_db]> SELECT * FROM categories;
+----+-----+
| id | name  |
+----+-----+
| 1  | Study |
| 2  | Work  |
| 3  | Personal |
| 4  | AWS   |
+----+-----+
4 rows in set (0.001 sec)
```

Вопрос: Отобразилась ли новая запись на реплике? Объясните почему.

Ответ: Да, новая запись появилась на реплике через несколько секунд. Read Replica автоматически синхронизируется с основной базой данных и копирует все изменения.

Вопрос: Для чего нужны Read Replicas?

Ответ: Read Replicas используются для:

- уменьшения нагрузки на основную базу;
- ускорения SELECT-запросов;
- масштабирования приложений;
- повышения отказоустойчивости; разделения операций чтения и записи.

Вопрос: Где это полезно?

Ответ: интернет-магазины; социальные сети; аналитические системы; приложения с большим количеством пользователей и запросов на чтение данных.

В таких системах: основная база выполняет INSERT/UPDATE/DELETE; реплики обрабатывают SELECT-запросы.

Шаг 6. Подключение приложения к базе данных

Разработала простое приложение на PHP, настроила его для развертывания. Код прикреплен к этому репозиторию

Вывод

В ходе лабораторной работы были получены практические навыки работы с облачными базами данных в Amazon Web Services. Была создана сетевая инфраструктура с использованием VPC, публичных и частных подсетей, а также настроены группы безопасности для безопасного взаимодействия между EC2-инстансом и базой данных.

В процессе работы был развернут экземпляр Amazon RDS MySQL, выполнено подключение к базе данных с виртуальной машины EC2 и реализованы базовые SQL-операции: создание таблиц, добавление записей, выполнение SELECT-запросов и JOIN. Также была изучена работа Read Replica, проверена репликация данных и особенности разделения операций чтения и записи.

Дополнительно было разработано и подключено простое PHP-приложение, работающее с базой данных Amazon RDS. В ходе выполнения лабораторной работы были закреплены навыки настройки облачной инфраструктуры, подключения сервисов AWS и организации безопасной работы с базами данных в облачной среде.

Веблиография

- Amazon Web Services
<https://aws.amazon.com/>
- Amazon RDS Documentation
<https://docs.aws.amazon.com/rds/>
- MySQL Documentation
<https://dev.mysql.com/doc/>